Problem Set VI, Macroeconomics III
University of Copenhagen

March 2026

Exercise 1

The early RBC literature made the strong assumption of a representative agent. In this exercise, you show that this assumption can be fairly easily relaxed, to account for consumers who do not optimze their consumption perfectly, but instead consume their entire labor income. Assume the economy is described by an RBC model. Profit maximizing firms maximize profits, their production function is Cobb-Douglas and takes capital k_t and labor hours h_t as inputs:

$$f(k_t, h_t) = z_t k_t^{\alpha} h_t^{1-\alpha}.$$

The firm rents capital from households at rate r_t and pays wage rate w_t per hour worked it uses.

The preferences of a fraction $(1 - \vartheta)$ of the population are:

$$U = \sum_{t=0}^{\infty} \beta^t u(c_t)$$
$$u(c_t) = \frac{1}{1-\sigma} c_t^{1-\sigma}; \quad \sigma \neq 1$$

These agents have the following budget constraint:

$$a_{t+1} + c_t = w_t h_t + (1 - \delta + r_t) a_t$$

They can borrow and save freely, their demand for assets must be equal to the level of capital in the economy. They are "unconstrained" in their decision making.

Lastly, a fraction ϑ of households are Hand-to-Mouth consumers. They simply eat their labor income:

$$c_t^{htm} = w_t h_t$$

They cannot access credit markets, they cannot borrow or save. Hence, they are "constrained" in their decision making. This behavior is not microfounded, it is just an assumption.

Both types of households supply labor inelastically, such that $h_t = 1$.

(a) Set up the firms problem and find expressions for the interest rates r_t and wages w_t as a function of k_t, h_t and z_t.

(b) Set up the maximization problem for the unconstrained households and derive their first order conditions. Show that their consumption path is described by an Euler equation.

(c) Assume the economy is in steady state, $z_t = 1 \forall t$. What is the level of capital in the economy? What are the levels of the wage rate and the interest rate?

(d) From the budget constraints of the two households, derive the aggregate resource constraint of the economy. There is no need to assume that $z_t = 1$, since the budget and resource constraints must hold even if $z_t \neq q$.

(e) Assume the economy is in the $z_t = 1$ steady state at the end of period $t = -1$. In period $t = 0$, productivity rises, such that $z_0 > 1$ in period 0, in period $t = 1$, $z_1 = 1$, again and the economy ends. In words, explain when aggregate consumption in period $t = 0$ rises more in response to this one-time-shock: when ϑ is high or when ϑ is low.

Exercise 2

Consider the following 2-period version of the Real Business Cycle (RBC) model: the representative consumer maximizes

$$\max_{c_0, c_1, h, a_1} \frac{c_0^{1-\sigma}}{1-\sigma} - B\frac{1}{1+\phi}h^{1+\phi} + \beta\frac{c_1^{1-\sigma}}{1-\sigma}$$

$$\text{subject to } c_0 + a_1 = hw_0$$

$$c_1 = (1+r)a_1$$

a) Set up the Lagrangian for this problem and derive the first order conditions. Explain the significance of the parameter σ. What does a high value of σ imply about the consumer's preferences?

There are two firms in the economy. Firm 0 operates only in period 0, and uses only labor. Firm 1 operates only in period 1, and uses only capital:

$$F^0(h) = z_0 h^{1-\alpha}; \quad \Pi^0 = F^0(h) - wh \tag{Firm 0}$$

$$y_1 = z_1 k^{\alpha}; \quad \Pi^1 = F^1(k) - rk \tag{Firm 1}$$

Each firm operates in a competitive market, taking prices as given.

b) Derive the first order conditions for the firm .

c) Impose market clearing onto the economy, such that all household savings are saved in firm capital and output in each period equals consumption. Show that h is independent of productivity z_0 and z_1. Why is this the case?

d) Derive an expression for consumption in period 0, c_0, only as a function of the model's parameters. Show that it takes the form:

$$c_0 = z_0 \xi \gamma^{\frac{1-\alpha}{1+\phi}}$$

Explain why consumption in period 0 is independent of productivity in period 1, z_1.